

Using Structural Similarity and Kolmogorov-Arnold Networks for Anatomical Embedding of Cortical Folding Patterns

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Code Repository



INTRODUCTION

Background

The 3-hinge gyrus (3HG) is a newly defined folding pattern, which is the conjunction of gyri coming from three directions in cortical folding. Many studies demonstrated that 3HGs can be reliable nodes when constructing brain networks or connectomes since they simultaneously possess commonality and individuality across different individual brains and populations.

Purpose

3HGs are identified and validated within individual spaces, making it difficult to directly serve as the brain network nodes due to the absence of cross-subject correspondence. The 3HG correspondences represent the intrinsic regulation of brain organizational architecture, traditional image-based registration methods tend to fail because individual anatomical properties need to be fully respected.

Contribution

To address this challenge, we propose a novel self-supervised framework for anatomical feature embedding of the 3HGs to build the correspondences among different brains. The core component of this framework is to construct a structural similarity-enhanced multi-hop feature encoding strategy based on the recently developed Kolmogorov-Arnold Network (KAN) for anatomical feature embedding.

METHODOLOGY

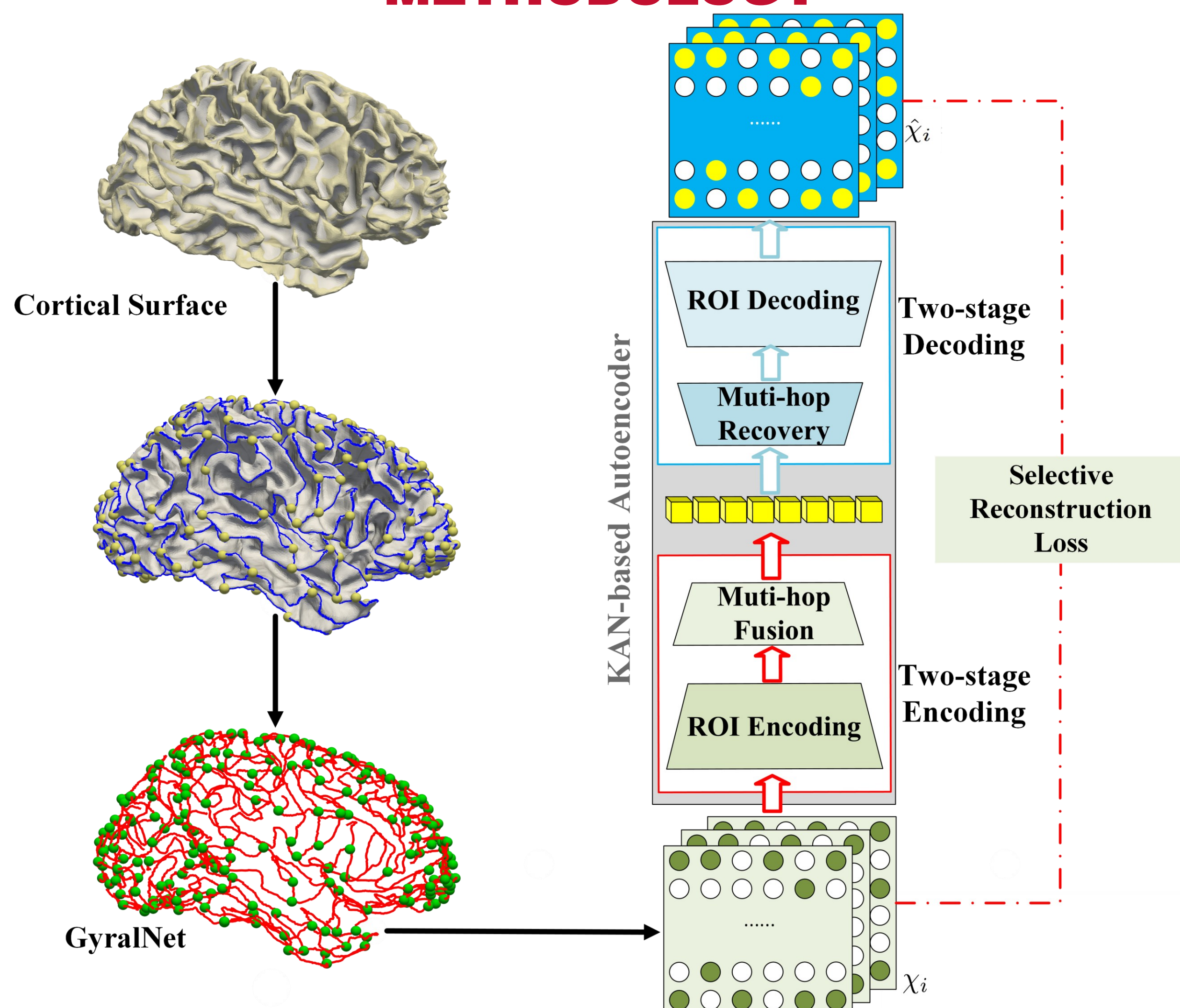


Fig. 1. Overall architecture of the proposed self-supervised framework. The framework constructs a structural similarity-enhanced multi-hop feature encoding strategy and uses recently proposed KAN for anatomical feature embedding of the 3HGs.

The structural similarity-enhanced hierarchical l -hop ($l > 0$) feature of the i -th 3HG node can be expressed as:

$$\chi_i = \begin{bmatrix} \langle S_0 \odot IF \rangle_i \\ \langle S_1 \odot A_1 F \rangle_i \\ \dots \\ \langle S_l \odot A_l F \rangle_i \end{bmatrix} \in \mathbb{R}^{(l+1) \times 75}$$

where $F \in \mathbb{R}^{n \times 75}$ is the one-hot ROI label of each 3HGs obtained from Destrieux atlas, A_k represents the k -hop adjacency matrix of GyratNet, $\langle \cdot \rangle_i$ denotes the i -row of the matrix and $\odot$ indicates the Hadamard product. S_k represents k -hop structural similarity, which is used to measure the similarity between two 3HGs. The self-supervised framework for anatomical feature embedding in 3HG employs a two-stage encoding process Φ that hierarchically maps structural similarity-enhanced multi-hop features to a latent representation θ and a two-stage decoding process $\hat{\Phi}$, which hierarchically reconstructs the original input from this latent representation. Each encoding and decoding process is implemented by a KAN layer, and the self-supervised framework f is a combination of four layers using the projection operator $\circ$:

$$\hat{\chi}_i = f(\chi_i) = (\hat{\Phi}_{ROI} \circ \hat{\Phi}_{MH} \circ \Phi_{MH} \circ \Phi_{ROI})\chi_i$$

Loss function for the i -th 3HG node:

$$\mathcal{L} = \|\hat{\chi}_i - \chi_i\|_2 + \|(\hat{\theta}_i - \theta_i) \odot (1_\theta + \lambda \theta_i)\|_2$$

RESULTS

In this study, we use structural MRI (T1-weighted) of 1064 subjects from the Human Connectome Project (HCP) S1200 release. The Destrieux Atlas was then applied for ROI labeling on the reconstructed white matter surface.

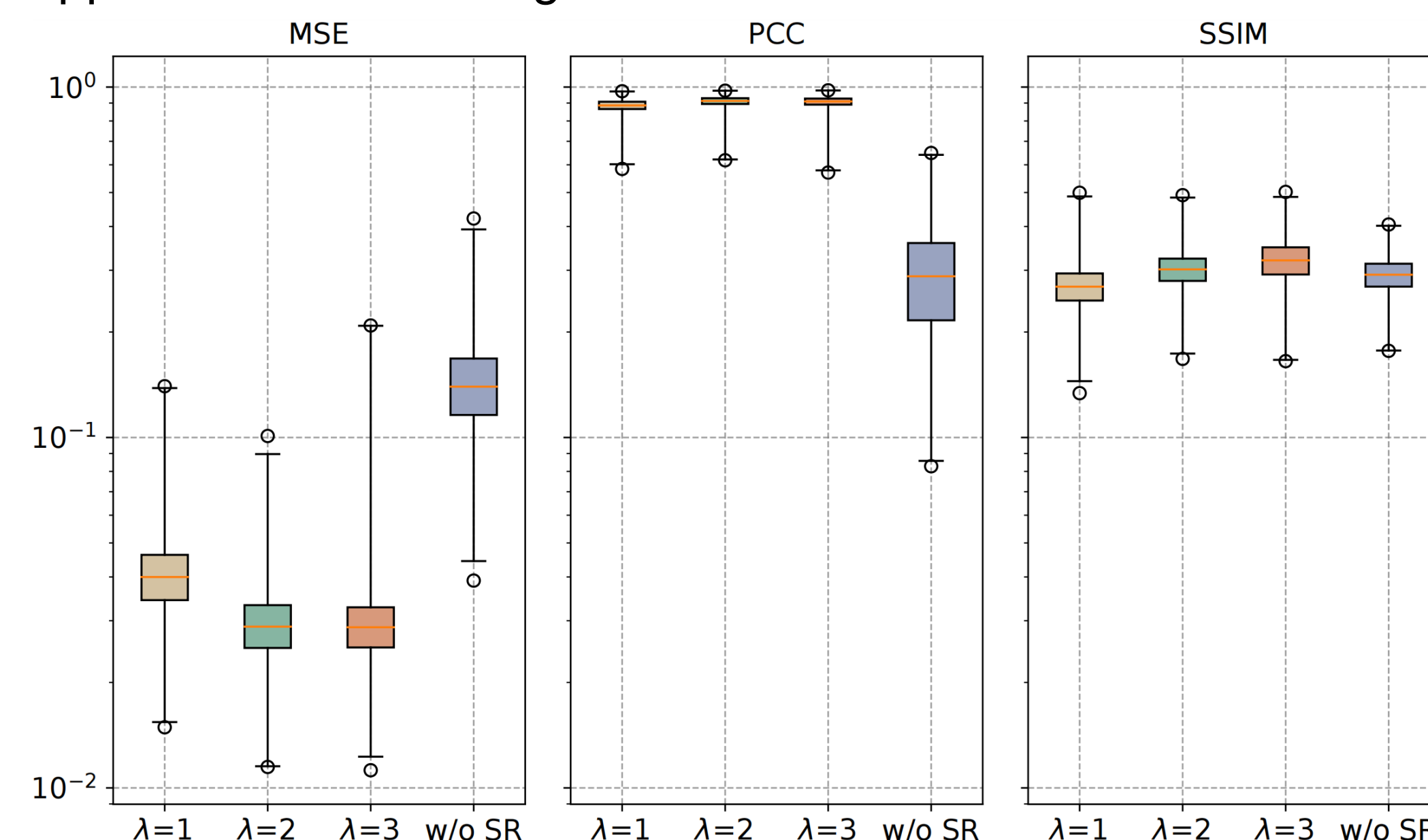


Fig. 2. Evaluation of different hyperparameter settings and ablation analysis of the proposed selective reconstruction loss (w/o SR) via MSE, PCC, and SSIM. Note that the y-axis is on a log scale.

Top-5 Connections from Learned ROI Embeddings

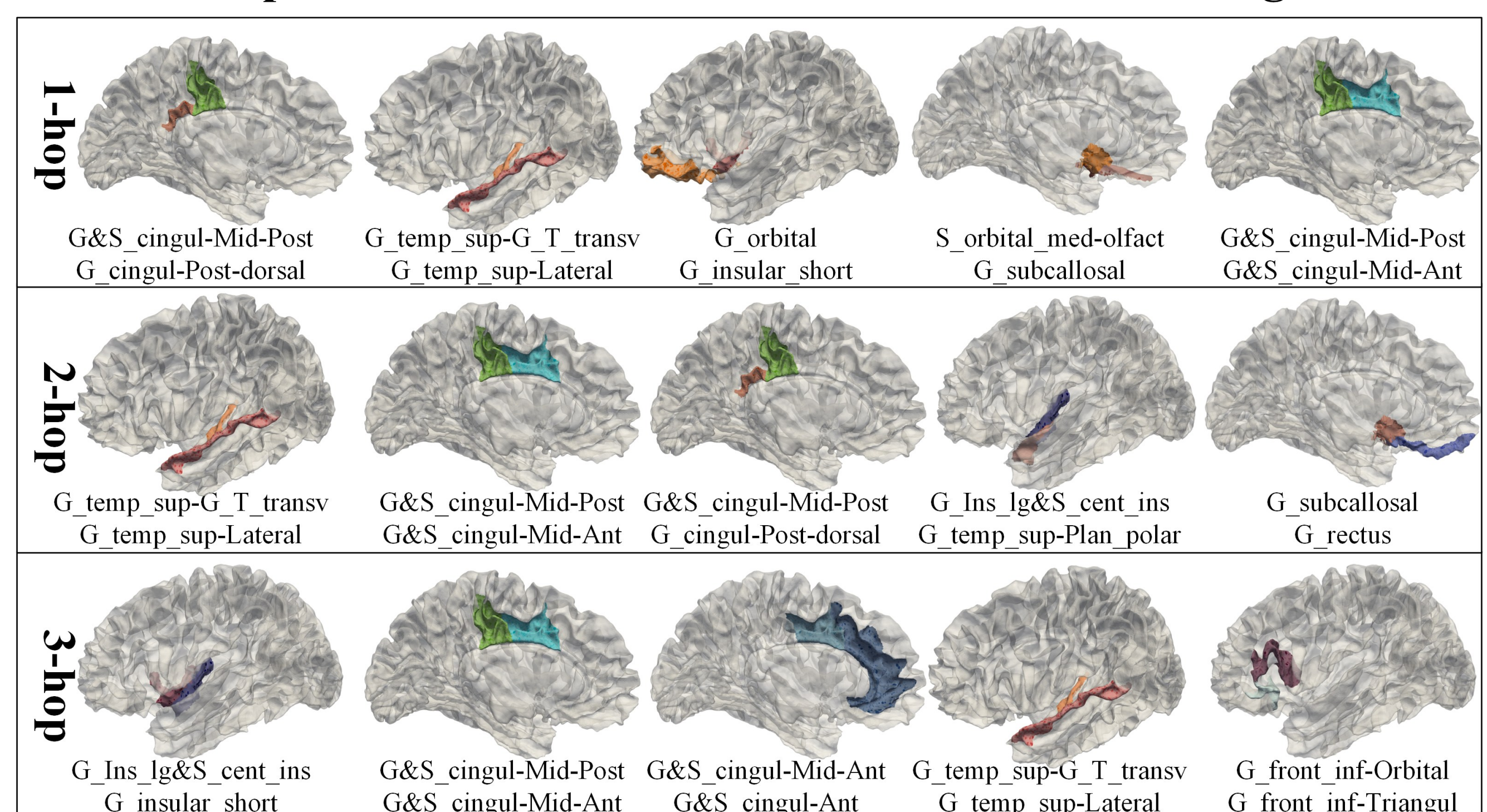


Fig. 3. ROI embedding connectivity. The Top-5 connections (PCC between pairs of ROIs) from the learned ROI embeddings. Within each row, connections are ordered by their rank from left to right. Color lookup table provided by Freesurfer.

Table 1. Quantitative results. ROI hit rate of the proposed method under several hop numbers and the baseline methods. The node with the highest cosine similarity is considered as the

Methods	ROI Hit Rate (%) $\uparrow$		
	L. Hemi.	R. Hemi.	Total
cortex2vec [12]	34.43	33.91	34.15
cortex2vec [12] +sse	34.58	33.80	34.17
ours (1-hop)	34.49	34.05	34.26
ours (2-hop)	34.70	33.82	34.24
ours (3-hop)	34.63	34.17	34.39

corresponding node to the anchor node. In most cases ($\approx 70\%$), no other 3HGs exhibit comparable cosine similarity, avoiding ambiguity in the matching process. 3HGs in the frontal and parietal lobe regions exhibit high variability, yet clear one-to-one correspondences can still be identified in these areas, while *cortex2vec* fails. However, "competing points"—indicated by red bubbles in the figure—may introduce potential one-to-many correspondences.

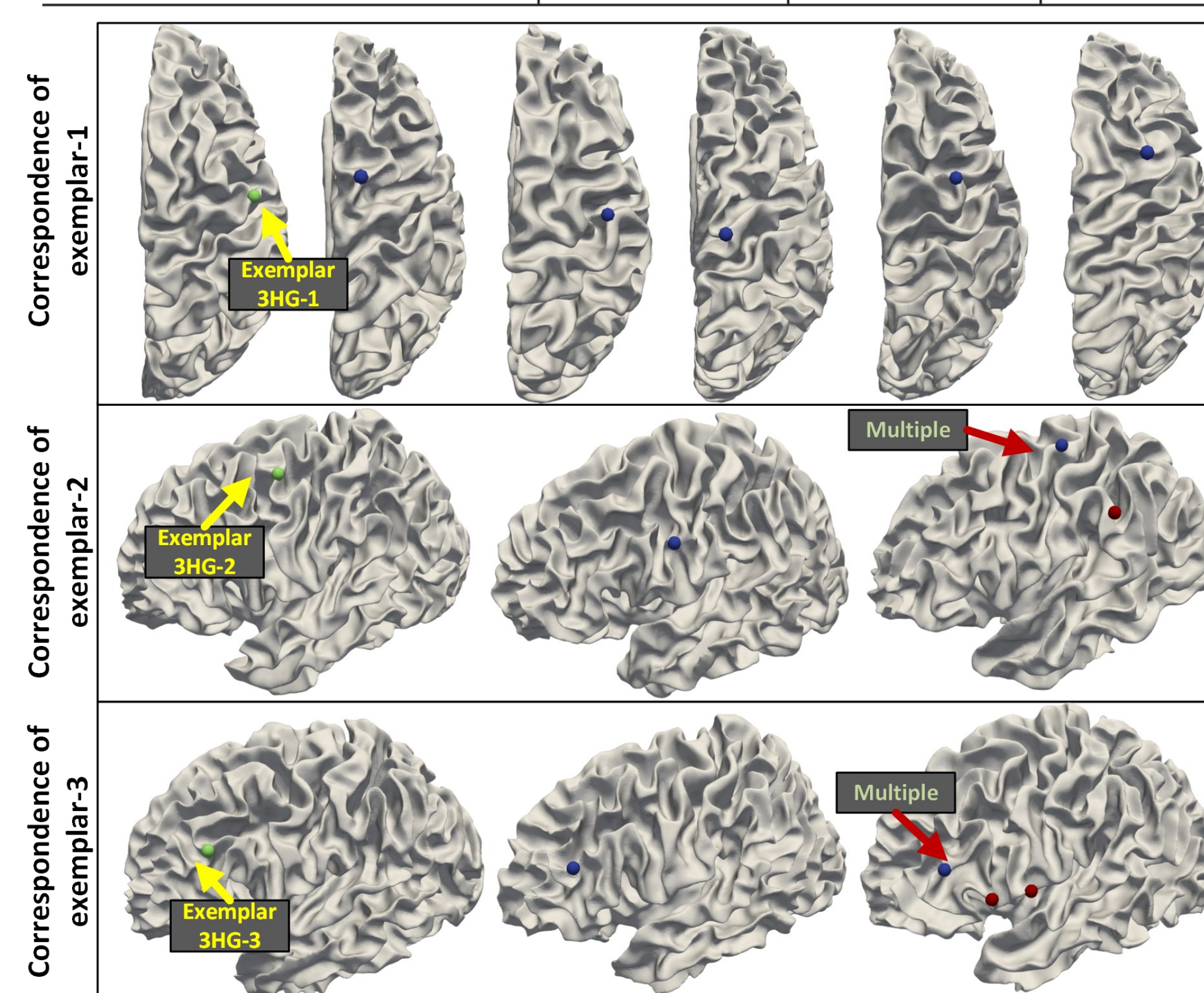


Fig. 4. Illustration of cross-subject correspondence. All 3HGs are represented as bubbles, with yellow indicating exemplars, blue indicating the identified corresponding points in other subjects, and red indicating the candidate points with cosine similarity close to the corresponding point.

ACKNOWLEDGEMENTS

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